



FIXWAALA: A HYPERLOCAL TRUST-GATED HOME-REPAIR SERVICE MARKETPLACE

by

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Specialization Project report submitted in partial fulfillment of
the requirements of IV trimester MCA,
CHRIST (Deemed to be University)

August 2026



CERTIFICATE

*This is to certify that the report titled **Fixwaala: A Hyperlocal Trust-gated Home-repair Service marketplace** is a bona fide record of work done by Jinisha Leema Rosario (2547122), Joshua George Philip (2547124) and Sovin Somy (2547157) of CHRIST(Deemed to be University), Bangalore, in partial fulfillment of the requirements of IV Trimester MCA during the year 2026.*

Project Guide

Head of the Department

Valued by:

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| | Date of Exam | : | |

AI USAGE DECLARATION

I hereby declare that Artificial Intelligence tools were used only for supportive purposes such as idea exploration, conceptual understanding, literature guidance, and grammar refinement during the development of this project. No AI generated code has been copied or submitted, and all programming, model development, experimentation, analysis, and results were independently developed and implemented by me or our team. I confirm that this project is our original work and reflects our own understanding and effort, and I take full responsibility for its authenticity and adherence to academic integrity policies.

I also declare that the approximate usage of Artificial Intelligence tools for supportive purposes in this project is 27%, limited to idea exploration, literature search guidance, grammar refinement, clarification of technical concepts, and other academic assistance.

The details of AI tools used in this project and their purpose are listed below:

1. Name of the AI Tool: Claude AI

Purpose of Use: Ideation and conceptual assistance

2. Name of the AI Tool: Gemini

Purpose of Use: Architecture guidance

3. Name of the AI Tool: Grammarly

Purpose of Use: Grammar and spell check

This declaration confirms that the AI tools mentioned above were used only for supportive and assistive purposes and that no AI generated code has been directly copied or submitted as part of this project.

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Project Title : Fixwaala: A Hyperlocal Trust-gated Home-repair Service marketplace

Signature :

Date : 18-08-2026

ACKNOWLEDGEMENT

In the pursuit of academic excellence and scholarly endeavors, I extend my sincere acknowledgments to those who have played pivotal roles in the realization of this project.

First and foremost, I offer my heartfelt gratitude to the Divine for His boundless grace and blessings, which have guided me throughout this journey.

I profoundly thank **Dr. Fr. Jossy P. George**, Director of Department of Computer Science, CHRIST (Deemed to be University), Bangalore, for providing me the opportunity to be a part of the institution. To **Dr. Rupali Sunil Wagh**, the Head, and **Dr. Gobi R**, the Associate Head of the Department of Computer Science at CHRIST (Deemed to be University), I express my profound appreciation for entrusting me with this project, which has been a significant part of my academic journey.

I extend my sincere gratitude to our Program Coordinator, **Dr. Cynthia T**, for her consistent academic support, guidance, and encouragement, which greatly ensured the successful completion of the project.

I extend my sincere gratitude to our Project Coordinator, **Dr. Kavitha R**, and Project Guide, **Dr. Tegil J. John**, for their constant guidance, invaluable mentorship, and insightful suggestions, which have played a significant role in shaping and refining this project.

I am deeply indebted to the distinguished faculty members of the Department of Computer Science at CHRIST (Deemed to be University) for their scholarly guidance, support, and expertise in the successful completion of this project.

Lastly, I express my sincere appreciation to my friends and beloved family members for their unwavering encouragement and support throughout this journey. Your belief in me has been a constant source of inspiration and motivation.

ABSTRACT

Fixwaala is a hyperlocal, on-demand home-repair marketplace that connects customers with nearby plumbers, electricians, and carpenters. Most existing platforms auto-assign a service provider through a backend algorithm, leaving the customer with little visibility or control over who enters their home. Fixwaala addresses this gap through a trust-gated matching mechanism: when a provider accepts a service ticket, the assignment is not finalised immediately. Instead, the customer is given a 30-second confirmation window to review the provider’s profile, verification status, ratings, and completed-job history before locking in the match, rejecting it, or allowing the window to time out and trigger a re-match. To further safeguard privacy, the customer’s exact address is withheld from the provider until this confirmation step is complete.

In addition to trust-gated matching, the platform incorporates guided, keyword-based fault classification at the point of ticket creation, with an AI/LLM-based classifier identified as a planned future enhancement. A dedicated provider dashboard offers visibility into earnings, acceptance and completion rates, rating trends, and a computed trust score, enabling providers to track and improve their performance over time. The system is implemented as a single Flutter codebase spanning the customer application, provider application, and admin panel (including its web build), and is backed by Firebase Authentication and Cloud Firestore, eliminating the need for a separate custom backend server.

This report documents the background, requirement analysis, and system design of Fixwaala in its current state, covering the existing-system landscape, the proposed system and its features, functional and non-functional requirements, system architecture, data flow, Firestore data design, and the interface and report design of the customer application, provider application, and admin panel. As this is an actively evolving student project, the report explicitly distinguishes between capabilities that are fully implemented, partially wired, and designed but not yet built, with a dedicated implementation-status section (3.8) provided so that the document remains an honest and accurate reference rather than an idealised account of the system.